Key Business Insights & Analytical Findings

- **The Overtime Burnout Multiplier:**
    - **Data:** Employees working overtime exhibit an attrition rate of **30.53%**, compared to just **10.44%** for non-overtime staff.
    - **Business Impact:** Overtime nearly triples turnover probability ($p < 10^{-20}$), accounting for **53.6% of all resignations** (127 out of 237) while representing only 28.3% of the total headcount.
- **Frontline Job Roles in the Danger Zone:**
    - **Data: Sales Representatives** suffer from the highest turnover at **39.76%** (33 departures out of 83), followed by **Laboratory Technicians** at **23.94%** (62 departures) and **Human Resources** at **23.08%** (12 departures).
    - **Business Impact:** High turnover in frontline roles disrupts core business execution and client touchpoints, driving continuous recruitment and onboarding costs.
- **Compensation Inequity & Early-Career Flight Risk:**
    - **Data:** Employees in the **Low Salary Tier** ($< \$3{,}500$/month) have an attrition rate of **27.12%**, compared to **11.31%** for Medium and **10.53%** for High earners.
    - **Business Impact:** The scatter plot confirms that turnover is concentrated heavily in the first 0–10 years of experience where compensation sits below $\$5$K--$\$10$K. Once employees cross this salary threshold, retention stabilizes.
- **Compounded Risk (Toxic Intersection of Low WLB & Low Satisfaction):**
    - **Data:** When poor work-life balance (**WorkLifeBalance = 1**) intersects with low job satisfaction (**JobSatisfaction = 1**), attrition surges to **47.06%**.
    - **Business Impact:** Poor work-life balance is an immediate flight trigger, accelerating turnover even among otherwise productive contributors.

2. Data Transformation & Power BI Implementation Details

- **Data Cleaning & Quality:**
    - The dataset contained **0 missing values** and **0 duplicate rows** across 1,470 records.
    - Two uninformative constant columns (Over18 and StandardHours) and EmployeeCount were removed to streamline the data model.
- **Calculated Columns (DAX):**
    - Attrition Numeric = IF('WA_Fn-UseC_-HR-Employee-Attriti'[Attrition] = "Yes", 1, 0)
    - Salary Tier = SWITCH(TRUE(), [MonthlyIncome] < 3500, "Low", [MonthlyIncome] <= 6500, "Medium", "High")
    - Salary Tier Order = SWITCH(TRUE(), [MonthlyIncome] < 3500, 1, [MonthlyIncome] <= 6500, 2, 3) *(Used for sorting Salary Tier logically)*
    - Experience Band = SWITCH(TRUE(), [TotalWorkingYears] <= 5, "Junior", [TotalWorkingYears] <= 10, "Mid-Level", [TotalWorkingYears] <= 20, "Experienced", "Senior")
    - Experience Band Order = SWITCH(TRUE(), [TotalWorkingYears] <= 5, 1, [TotalWorkingYears] <= 10, 2, [TotalWorkingYears] <= 20, 3, 4)
    - Age Group = SWITCH(TRUE(), [Age] < 30, "Under 30", [Age] <= 40, "30 - 40", [Age] <= 50, "41 - 50", "Above 50")
    - Age Group Order = SWITCH(TRUE(), [Age] < 30, 1, [Age] <= 40, 2, [Age] <= 50, 3, 4)
- **Core DAX Measures:**
    - Total Employees = COUNTROWS('WA_Fn-UseC_-HR-Employee-Attriti')
    - Employees Left = CALCULATE(COUNTROWS('WA_Fn-UseC_-HR-Employee-Attriti'), 'WA_Fn-UseC_-HR-Employee-Attriti'[Attrition] = "Yes")
    - Employees Stayed = CALCULATE(COUNTROWS('WA_Fn-UseC_-HR-Employee-Attriti'), 'WA_Fn-UseC_-HR-Employee-Attriti'[Attrition] = "No")
    - Attrition Rate = DIVIDE([Employees Left], [Total Employees], 0)
    - Avg Monthly Income = AVERAGE('WA_Fn-UseC_-HR-Employee-Attriti'[MonthlyIncome])
    - OverTime Attrition Rate = CALCULATE([Attrition Rate], 'WA_Fn-UseC_-HR-Employee-Attriti'[OverTime] = "Yes")
    - Non-OverTime Attrition Rate = CALCULATE([Attrition Rate], 'WA_Fn-UseC_-HR-Employee-Attriti'[OverTime] = "No")
    - Sales Rep Turnover = CALCULATE([Attrition Rate], 'WA_Fn-UseC_-HR-Employee-Attriti'[JobRole] = "Sales Representative")
    - Low Salary Tier Turnover = CALCULATE([Attrition Rate], 'WA_Fn-UseC_-HR-Employee-Attriti'[Salary Tier] = "Low")